

1905 Einstein: Zur Elektrodynamik bewegter Körper (§7 aberration) — verification companion

Albert Einstein

Einstein 1905 §7 — verification companion

[!abstract] Legend - □ tip/note — what the source establishes and how it is used - □ quote — verbatim source text

[!danger] PDF not in this folder Add 1905_Einstein_Elektrodynamik_bewegter_Koerper.pdf to project_sources/ to verify §7 directly. Only the citation and the §7 formula are used below; no page number is asserted without the PDF.

Bibliographic Info

- **Author:** Albert Einstein
- **Title:** Zur Elektrodynamik bewegter Körper
- **Year:** 1905
- **Publication:** Annalen der Physik, vol. 17, pp. 891-921
- **Section used:** §7, “Theorie der Aberration und des Dopplerprinzips”

Claim used in the sheet / notebook

[!tip] Relativistic aberration formula (§7) $\cos \varphi' = (\cos \varphi - \beta) / (1 - \beta \cos \varphi)$ where φ is the ray direction in one frame, φ' in the frame moving at β . To first order this is $\Delta\varphi \approx \beta \sin \varphi$.

[!note] How it feeds the AIR block The formula returns **zero aberration when source and observer share a rest frame** ($\beta = 0$ between them), and β when they are in relative motion. That is exactly the corrected AIR layout: $0''$ in the $v=0$ columns, $\arctan(\beta) = 20.5''$ in the $v=\pm\beta$ columns. It is the independent (non-Franco) authority for “rest frame $\rightarrow 0$, moving frame $\rightarrow \beta$,” matching Franco’s slide 3 eq (7) for $n = 1$.

[!warning] Verify against the PDF Confirm the formula sits in §7 and that Einstein states the $\theta = 0$ (source-rest) limit. The 1923 Methuen English translation (“On the Electrodynamics of Moving Bodies”) carries the same §7 numbering.

Backlinks

- [[2025_Ligabue_Airy_Relativity]]

Tags

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